BookNest: Software Requirements Specification

Version 0.1 (Draft) · Status: for review

1. Introduction

1.1 Purpose

This document defines what BookNest must do and the qualities it must have. It is the reference for design, development, and testing.

1.2 Product vision

BookNest is a web application where readers manage a personal digital library, track their reading, and connect with other readers. It should feel like a cozy physical bookstore: warm, elegant, and calm.

1.3 Scope

BookNest has three parts:

- **My Library:** books, collections, progress, quotes, notes, goals.
- **Reading Community:** profiles, follows, reviews, likes, comments, recommendations.
- **Reading Insights:** dashboard, streaks, statistics, charts.

Out of scope for version 1: native mobile apps, payments or book sales, e-book reading or file hosting, AI recommendations, direct messaging, and multi-language support.

1.4 Definitions

Term	Meaning
Book	Shared metadata record (title, author, ISBN, pages, cover).
User book	One user's relationship to a book (status, progress, rating).
Shelf / Collection	A user-created group of books.

Term	Meaning
Reading log	One record of pages read on a given day.
Streak	Consecutive days with at least one reading log.
Visibility	`private`, `followers`, or `public`.

2. Overall description

2.1 User roles

- **Guest:** can view the landing page and public reviews, and register or log in.
- **Reader (registered user):** full access to their own library, tracker, insights, and social features.
- **Admin:** moderates reported content and manages users and the shared book catalog.

2.2 Operating environment

Modern desktop and mobile browsers. The app is a React single-page frontend talking to a Node.js/Express REST API backed by MySQL.

2.3 Assumptions and constraints

- Book metadata comes from the Google Books API; the app must still work for manual entries if the API is unavailable or rate-limited.
- Cover images are stored in Cloudinary; the database stores only URLs.
- Authentication uses JWT.
- The project is built by one developer in phases, so every feature must be deliverable independently.

3. Functional requirements

Priority: **M** = must have, **S** = should have, **C** = could have.

3.1 Authentication and profiles

ID	Requirement	Priority
FR-1	A guest can register with username, email, and password. Username and email must be unique, and the email must be verified before first login.	M
FR-2	A user can log in and log out; sessions use JWT access tokens.	M
FR-3	Passwords are stored hashed (bcrypt or argon2), never in plain text.	M
FR-4	A user can edit their profile: display name, bio, avatar, favorite genres.	M
FR-5	A user can reset a forgotten password by email.	S
FR-6	A user can choose whether their profile and activity are public or private.	S
FR-7	A user can delete their account and all associated data.	S

3.2 My Library

ID	Requirement	Priority
FR-8	A user can add a book manually (title, author, pages, genre, optional ISBN).	M
FR-9	A user can search Google Books by title, author, or ISBN and add a result with its details prefilled.	M
FR-10	A user can upload a cover image (JPG/PNG/WebP, max 5 MB) or use the cover from Google Books.	M
FR-11	A user can set a book's status: Want to Read, Currently Reading, or Completed.	M
FR-12	A user can create, rename, reorder, and delete collections, and assign books to them.	M

ID	Requirement	Priority
FR-13	A user can search and filter their library by title, author, status, collection, genre, and rating.	M
FR-14	A user can view their books on a visual bookshelf and drag books to rearrange them.	S
FR-15	A user can edit or remove a book from their library. Removing it deletes their notes, logs, and review for that book.	M
FR-16	The system avoids duplicate catalog entries (matching on ISBN or Google Books ID) and reuses the existing record.	S

3.3 Reading tracker

ID	Requirement	Priority
FR-17	A user can record their current page for a book they are reading.	M
FR-18	A user can add a reading log (pages read, date) and the system updates progress.	M
FR-19	The system shows progress as a percentage of total pages.	M
FR-20	When progress reaches the last page, the system offers to mark the book Completed and records the finish date.	S
FR-21	A user can set an annual reading goal (number of books) and see progress toward it.	M
FR-22	The system calculates the current and longest reading streak.	M
FR-23	A user can set an optional daily page or minute goal.	C

3.4 Reviews, quotes, and notes

ID	Requirement	Priority
FR-24	A user can rate a book from 1 to 5 stars.	M
FR-25	A user can write, edit, and delete a review for a book.	M
FR-26	A user can save quotes with an optional page number.	M
FR-27	A user can write private notes on a book.	M
FR-28	Reviews and quotes have a visibility setting; the default is private. Notes are always private.	M

3.5 Reading community

ID	Requirement	Priority
FR-29	A user can follow and unfollow other users freely, with no approval needed.	M
FR-30	A user can view other users' public profiles, showing their public reviews, favorite books, and current reads.	M
FR-31	A user can like and unlike a review.	M
FR-32	A user can comment on a review and delete their own comments. Comments are flat, not threaded.	M
FR-33	A user can recommend a book to a follower with a short message.	S
FR-34	The home feed shows recent public activity from followed users (reviews, completed books, recommendations).	S
FR-35	A user can report a review or comment for moderation.	S

3.6 Discover

ID	Requirement	Priority
FR-36	A user can browse books by genre or category.	M
FR-37	The system shows popular books, ranked by how many users have added them and by average rating.	S
FR-38	A book's page shows its details, average rating, and public reviews from other readers.	M
FR-39	The system suggests books based on the user's favorite genres and highly rated books, using simple rule-based logic.	C

3.7 Insights dashboard

ID	Requirement	Priority
FR-40	The dashboard shows total books read, books currently reading, total pages read, and current streak.	M
FR-41	The dashboard shows annual goal progress.	M
FR-42	A chart shows pages read and books completed per month.	M
FR-43	A chart shows the user's favorite genres.	S
FR-44	The user can choose which year's statistics to view.	C

3.8 Administration

ID	Requirement	Priority
FR-45	An admin can view reported content and remove it.	S
FR-46	An admin can suspend or reactivate a user account.	S

ID	Requirement	Priority
FR-47	An admin can edit or merge duplicate catalog books.	C

4. Non-functional requirements

4.1 Security

- **NFR-1:** All traffic uses HTTPS in production.
- **NFR-2:** Every API route that returns or changes user data verifies the JWT and checks ownership or visibility rules.
- **NFR-3:** All SQL uses parameterized queries.
- **NFR-4:** User-generated text is sanitized or escaped to prevent XSS.
- **NFR-5:** Login and registration endpoints are rate-limited.
- **NFR-6:** Uploads are validated for type and size before being sent to Cloudinary.
- **NFR-7:** Secrets (database credentials, JWT secret, Cloudinary keys) live in environment variables and never in the repository.

4.2 Performance

- **NFR-8:** Typical API responses return in under 500 ms under normal load.
- **NFR-9:** List endpoints (library, feed, reviews) are paginated.
- **NFR-10:** Cover images are served in optimized sizes through Cloudinary transformations and lazy-loaded.

4.3 Usability and design

- **NFR-11:** The interface is fully responsive from 360 px phones to desktop.
- **NFR-12:** The visual style uses a warm, cozy palette, elegant serif typography for headings, and smooth, subtle animations.
- **NFR-13:** Core flows (add a book, log reading) take no more than three steps.
- **NFR-14:** The app meets basic accessibility: keyboard navigation, sufficient color contrast, alt text on covers, and respect for reduced-motion settings.

4.4 Reliability and data integrity

- **NFR-15:** The database enforces foreign keys and unique constraints.
- **NFR-16:** If Google Books or Cloudinary is unavailable, the user sees a clear message and can continue with manual entry.
- **NFR-17:** Daily database backups in production.

4.5 Maintainability

- **NFR-18:** The API follows REST conventions and is documented (OpenAPI or a README).
- **NFR-19:** The codebase is split into routes, controllers, services, and models, with schema changes tracked as migrations.
- **NFR-20:** Core business logic (streaks, progress, visibility checks) has automated tests.

5. Key use cases

UC-1: Add a book from search

Actor: Reader · **Precondition:** logged in.

1. The reader opens Add Book and searches for a title.
2. The system lists matching Google Books results.
3. The reader selects one; the form is prefilled.
4. The reader picks a status and optional collection, then saves.
5. The system stores the catalog record (if new) and links it to the reader.

Alternative: No result found, so the reader fills in the form manually and uploads a cover.

UC-2: Log a reading session

Actor: Reader · **Precondition:** the book is Currently Reading.

1. The reader opens the book and chooses Log Reading.
2. The reader enters pages read (or the new current page) and the date.
3. The system saves the log, updates progress, and recalculates the streak.
4. If the last page is reached, the system offers to mark the book Completed.

UC-3: Publish a review

Actor: Reader · **Precondition:** the book is in the reader's library.

1. The reader rates the book and writes a review.
2. The reader sets visibility to public.
3. The system publishes it to the book page, the reader's profile, and followers' feeds.

UC-4: Follow a reader and interact

Actor: Reader.

1. The reader finds another user through search or a review.
2. The reader follows them.
3. Their public reviews appear in the reader's feed, where the reader can like or comment.

6. Data privacy rules

- New reviews and quotes default to **private**.
- Notes are never visible to anyone but their owner.
- Reading logs and statistics are private unless the user opts in to showing them on their profile.
- Email addresses are never shown to other users.
- Deleting an account removes the user's reviews, comments, likes, notes, logs, and follows.

7. Release plan

Release	Contents	Maps to roadmap phases
MVP	FR-1 to 4, 8 to 13, 15, 17 to 19, 21, 22, 24 to 28, 40 to 42	3, 4, 5, 7 (core)
Release 2	FR-14, 16, 20, 29 to 32, 36, 38	4, 6
Release 3	FR-5 to 7, 33 to 35, 37, 43, 45, 46	6, 7, 8
Later	FR-23, 39, 44, 47	after launch

8. Acceptance criteria for the MVP

- A new user can register, log in, add a book (by search and manually), and see it in their library.
- A user can log reading, see progress update, and see a correct streak.
- A user can rate and review a book, and private content is not reachable by any other account.
- The dashboard displays correct totals and a monthly chart.
- The app is usable on a phone and deployed at a public URL.

9. Decisions

1. Visibility is private or public only; there is no followers-only option.
2. Following is open; no approval step.
3. Email verification is required at registration.
4. Comments are flat, not threaded.
5. A book can appear only once in a user's library, so there are no duplicates. This means one row per user and book in the `user_books` design.